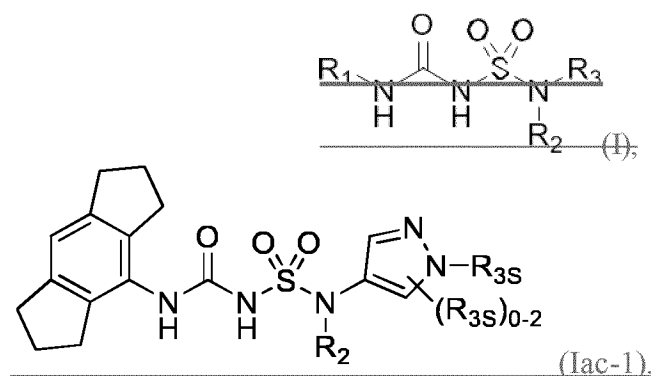


AUXILIARY REQUEST 3

CLAIMS:

1. A compound of Formula (Iac1):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

~~R₁ is C₃-C₁₆ cycloalkyl optionally substituted by one or more R_{1s}, wherein the C₃-C₁₆ cycloalkyl is a C₃-C₈ monocyclic cycloalkyl or polycyclic cycloalkyl, or, R₁ is a phenyl substituted by two or three R_{1s}, wherein each R_{1s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo, and at least one R_{1s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;~~

~~R₂ is -(CHX₂X₂)_n-R_{2s}, wherein n is 0, 1, or 2, and each X₂ is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;~~

~~R_{2s} is unsubstituted 5- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and~~

~~R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more R_{3s}, wherein each R_{3s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.~~

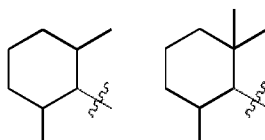
2. The compound of claim 1, wherein ~~R₁ is C₂-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic~~

cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or phenyl, wherein the C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, or C₁₂-C₁₆ tricyclic cycloalkyl is optionally substituted by one or more R_{1s}, the phenyl is substituted by two or three R_{1s}, and wherein at least one R_{1s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;

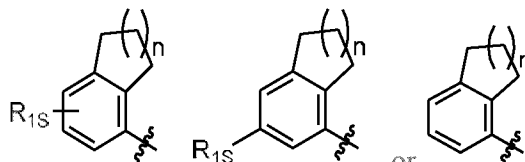
optionally wherein:

(a) the C₃-C₇ monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more R_{1s}; further

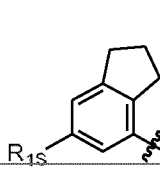
optionally wherein the C₃-C₇ monocyclic cycloalkyl is



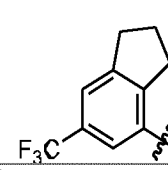
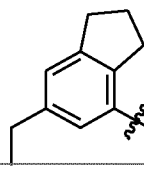
(b) the C₉-C₁₀ bicyclic cycloalkyl is



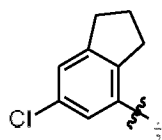
is 0, 1, or 2; optionally



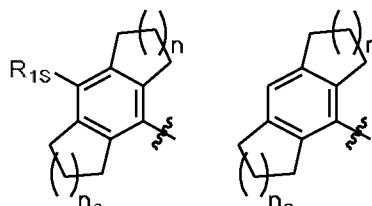
; further optionally



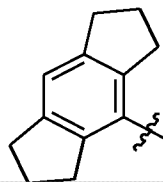
, or



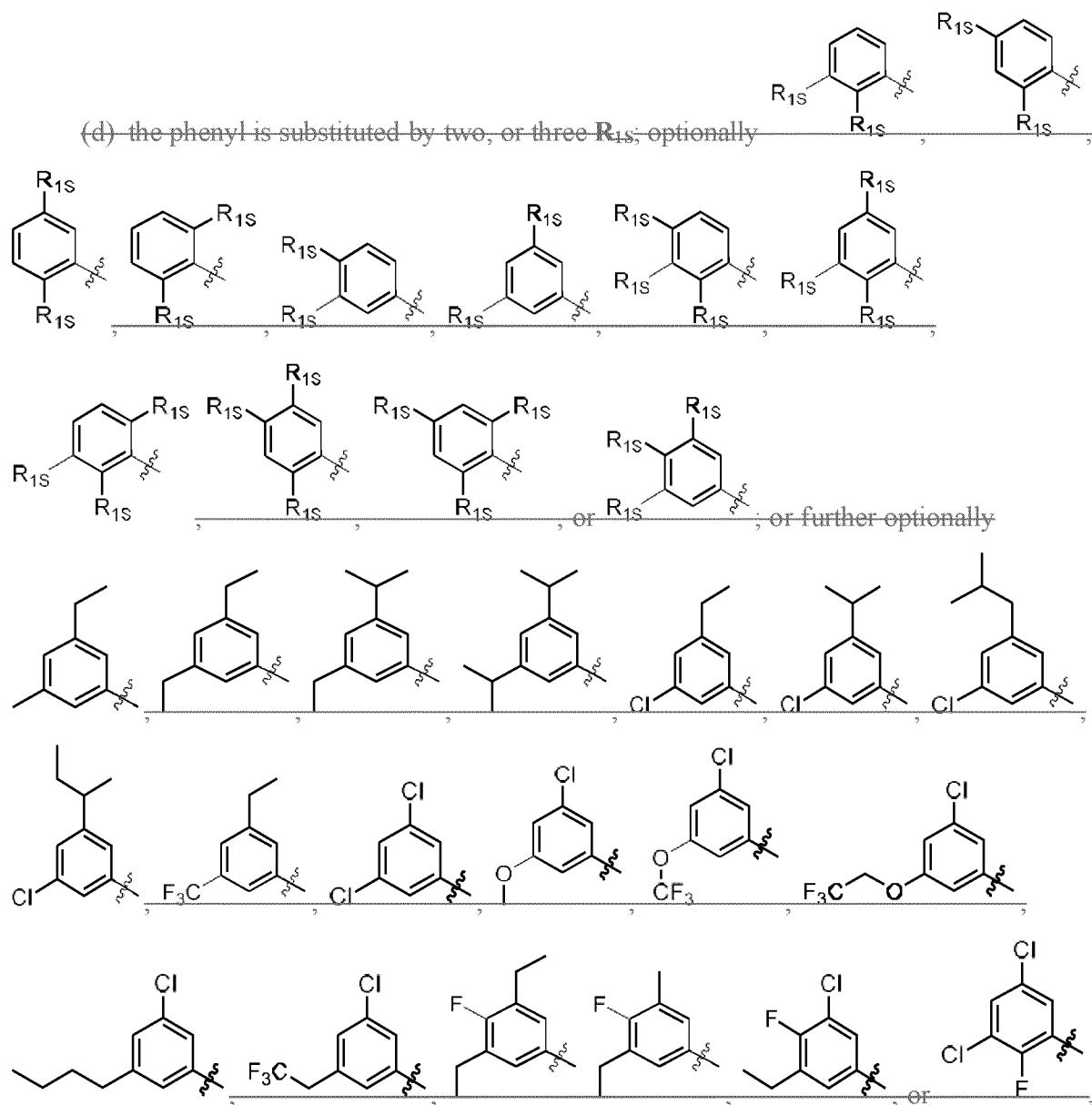
(c) the C₁₂-C₁₆ tricyclic cycloalkyl is



each independently are 0, 1, 2, or 3; further optionally



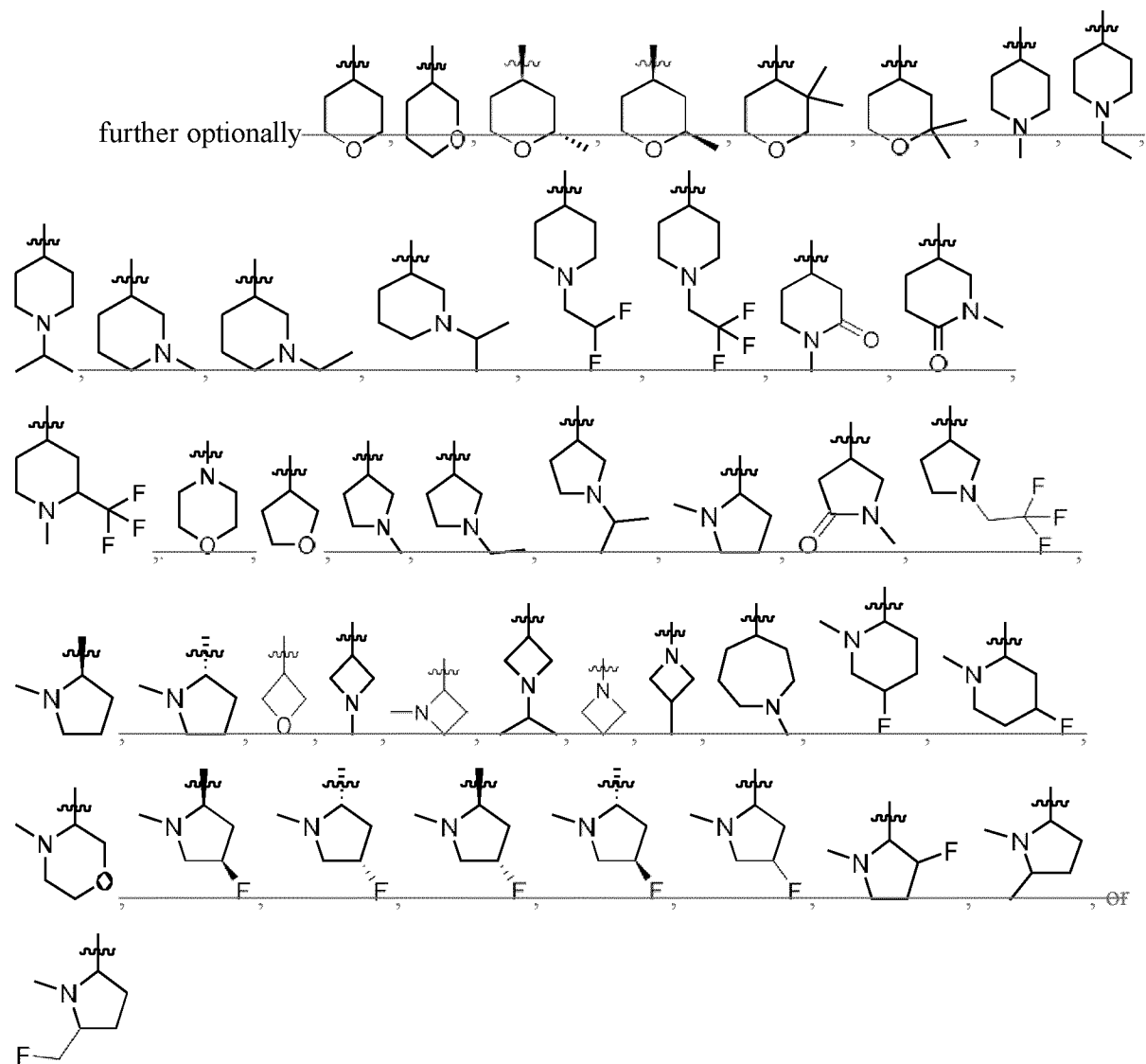
; and/or



3. ~~The compound of claim 1 or 2, wherein R_2 is R_{2s} , (CX_2X_2) R_{2s} , or $(CX_2X_2)_2$ R_{2s} .~~

42. ~~The compound of any one of claims 1 to 3, wherein R_{2s} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C_1 - C_6 alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, C_1 - C_6 haloalkyl, halo, CN, OH, $O(C_1$ - C_6 alkyl), NH_2 , $NH(C_1$ - C_6 alkyl), $N(C_1$ - C_6 alkyl) $_2$, or oxo;~~

Optionally oxetanyl, azetidiny, tetrahydrofuranyl or pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl is optionally substituted with one or more C_1 - C_6 alkyl, C_1 - C_6 haloalkyl, or oxo; or

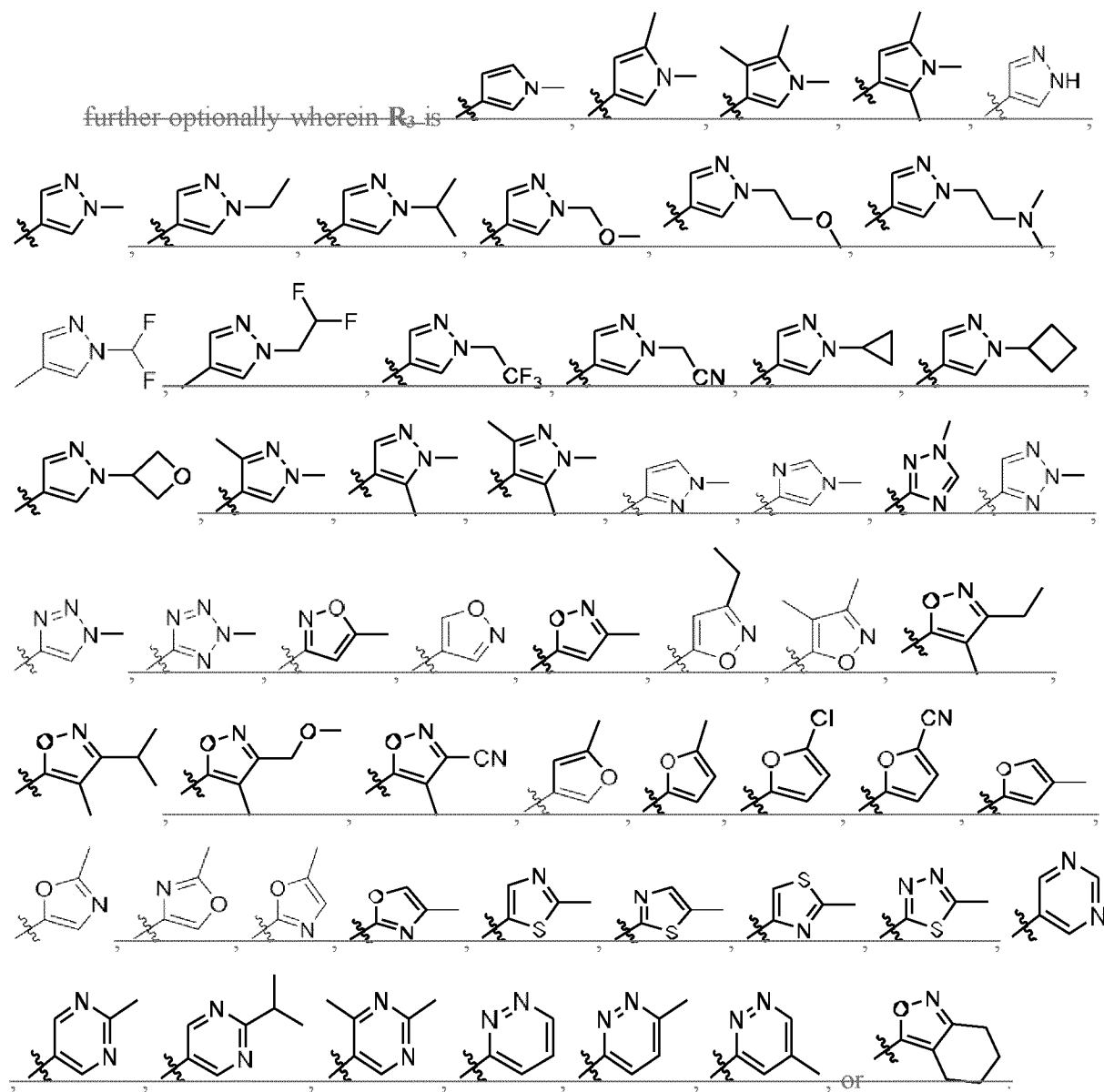


5. The compound of any one of claims 1 to 4, wherein R_3 is 7- to 12-membered heterocycloalkyl optionally substituted with one or more R_{3a} ;

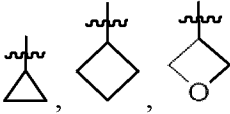
optionally wherein R_3 is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-

tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more R_{3s} ; or

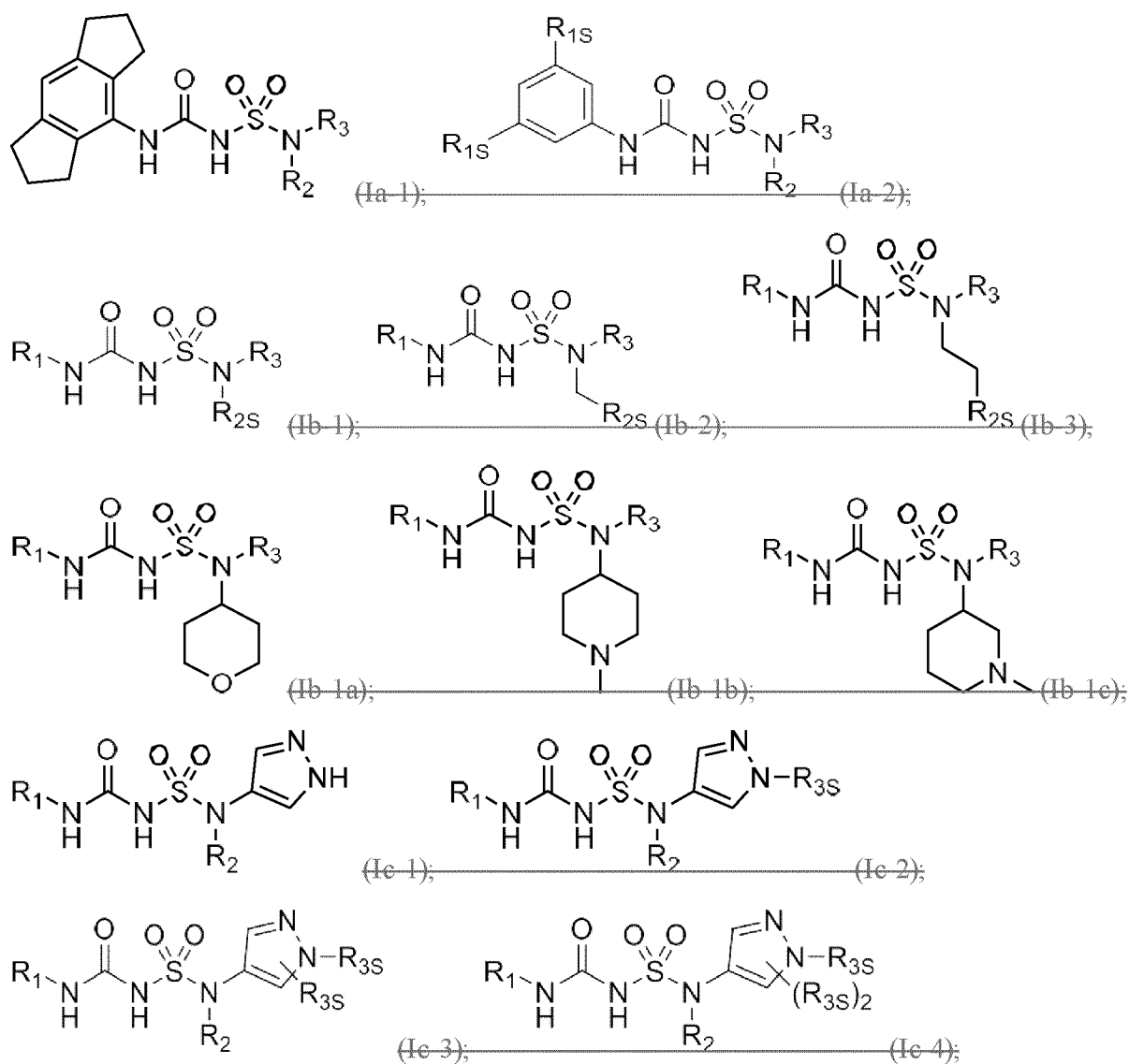
~~further optionally wherein R₃ is~~

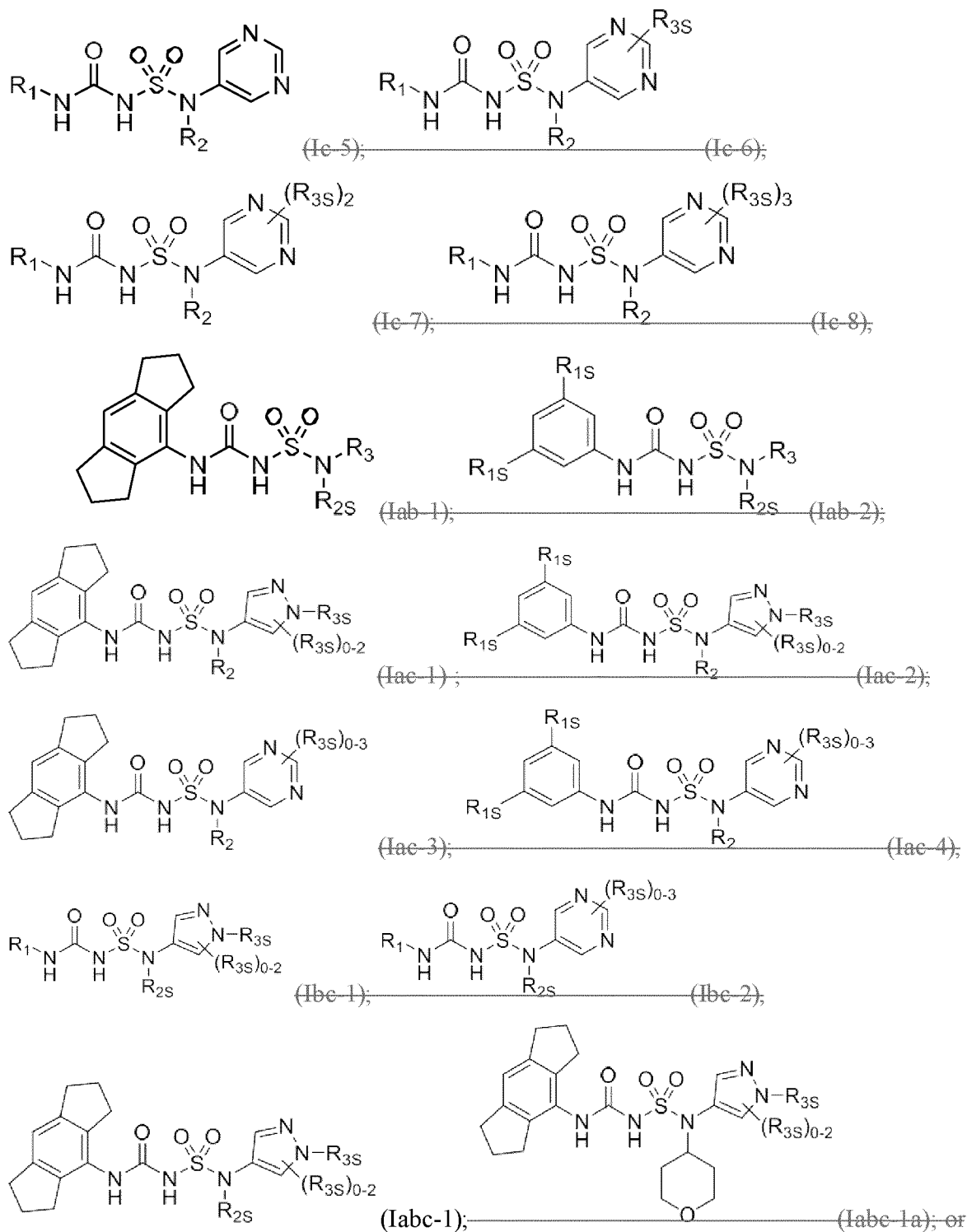


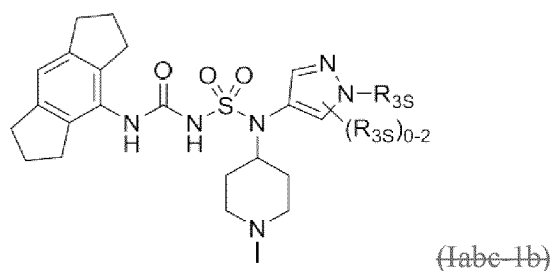
36. The compound of ~~any one of claims 1 or claim 2~~ to 5, wherein at least one **R**_{3S} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN;

optionally wherein at least one R_{3s} is , methyl, ethyl, isopropyl, -CH₂OCH₃, -CH₂CF₃, -CH₂CH₂OCH₃, -CH₂CN, CH₂CF₂, or -CH₂CH₂N(CH₃)₂.

74. The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Ia-1)~~, ~~(Ia-2)~~, ~~(Ib-1)~~, ~~(Ib-2)~~, ~~(Ib-3)~~, ~~(Ib-1a)~~, ~~(Ib-1b)~~, ~~(Ib-1c)~~, ~~(Ic-1)~~, ~~(Ic-2)~~, ~~(Ic-3)~~, ~~(Ic-4)~~, ~~(Ic-5)~~, ~~(Ic-6)~~, ~~(Ic-7)~~, ~~(Ic-8)~~, ~~(Iab-1)~~, ~~(Iab-2)~~, ~~(Iac-1)~~, ~~(Iac-2)~~, ~~(Iac-3)~~, ~~(Iac-4)~~, ~~(Ibc-1)~~, ~~(Ibc-2)~~, ~~(Iabc-1)~~, ~~(Iabc-1a)~~, or ~~(Iabc-1b)~~:

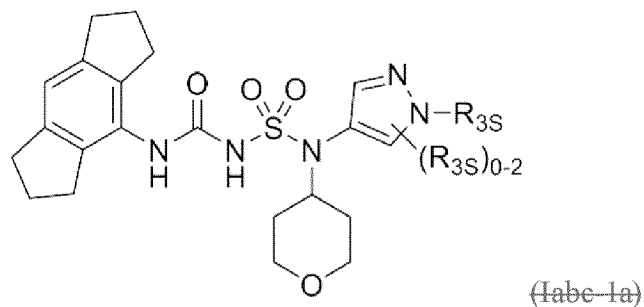






or a solvate, or pharmaceutically acceptable salt thereof, wherein ~~R₁~~, ~~R_{2S}~~, ~~R₂~~, and ~~R_{3S}~~ are as described ~~herein~~ in claim 1.

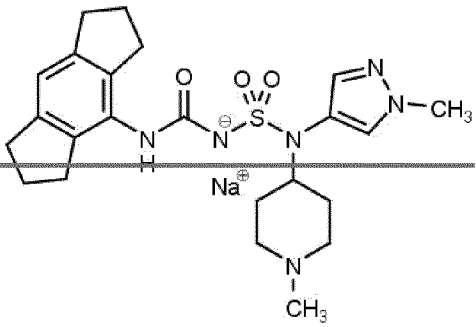
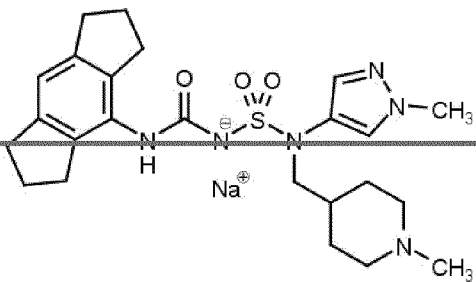
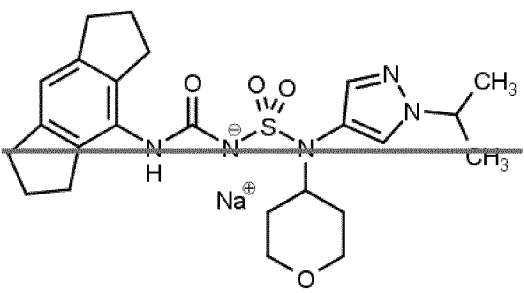
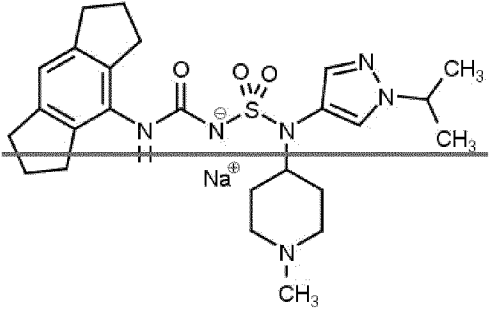
~~8. The compound of any one of claims 1 to 7, wherein the compound is of Formula (Iabc-1a):~~

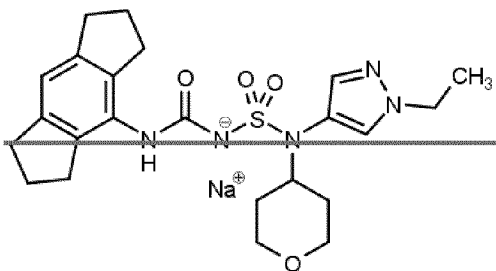
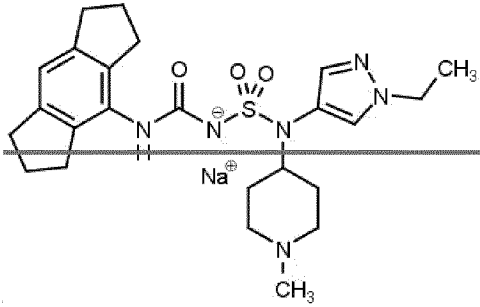
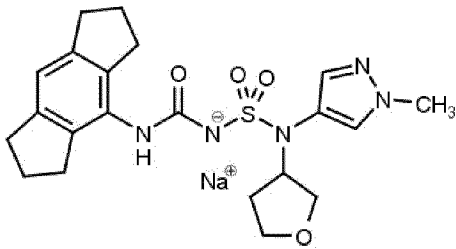
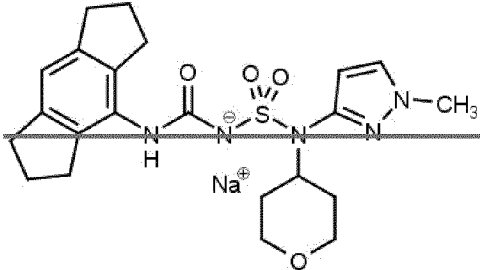
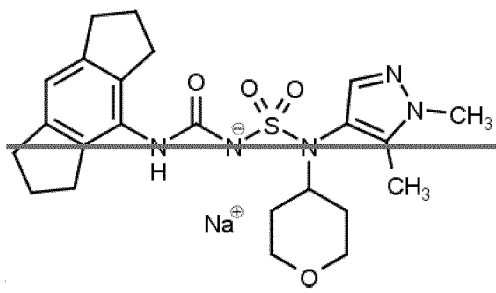


~~or a solvate, or pharmaceutically acceptable salt thereof, wherein ~~R_{3S}~~ is as described herein.~~

~~95. The compound of any one of the claims 1 to 84, being selected from:~~

Compound No.	Structure
1	

Compound No.	Structure
2	
3	
4	
5	

Compound No.	Structure
6	
7	
8	
9	
10	

Compound No.	Structure
11	 <chem>Cc1nc(C)c(C2=CC=C(C3=CC=CC=C3C4=CC=CC=C4)C(=O)N2S(=O)(=O)N1)C5=CC=CC=C5C6=CC=CC=C6</chem>
12	 <chem>Cc1cc2oc(C)nn2c1N(S(=O)(=O)N3=CC=CC=C3C4=CC=CC=C4)C5=CC=CC=C5C6=CC=CC=C6</chem>
13	 <chem>c1cc2oc(C)nn2c1N(S(=O)(=O)N3=CC=CC=C3C4=CC=CC=C4)C5=CC=CC=C5C6=CC=CC=C6</chem>
14	 <chem>Cc1nc(C)c(C2=CC=C(C3=CC=CC=C3C4=CC=CC=C4)C(=O)N2S(=O)(=O)N1)C5=CC=CC=C5C6=CC=CC=C6</chem>

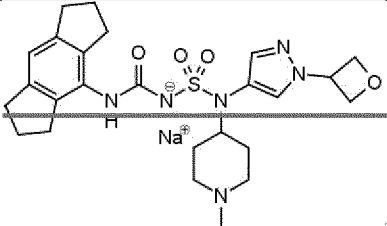
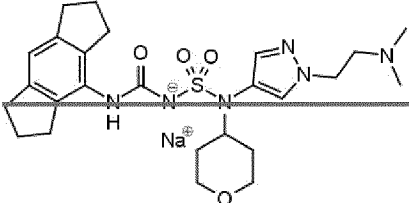
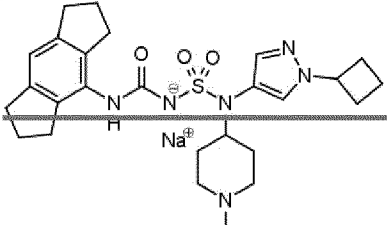
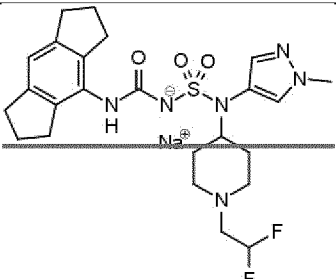
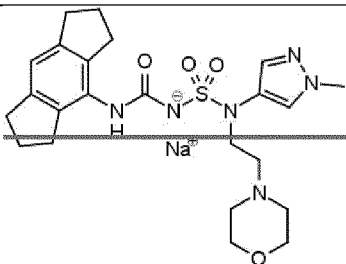
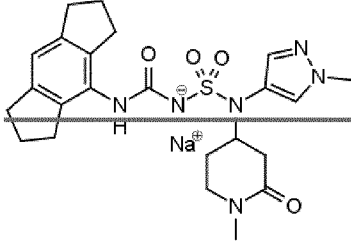
Compound No.	Structure
15	
16	
17	
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19	

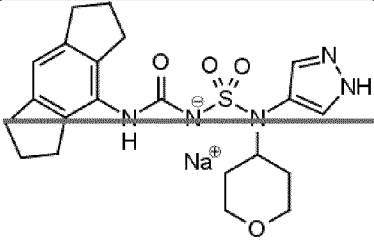
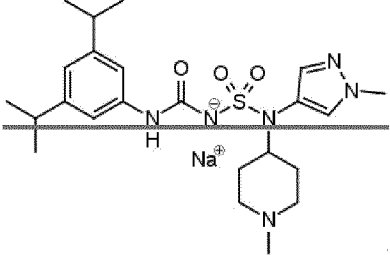
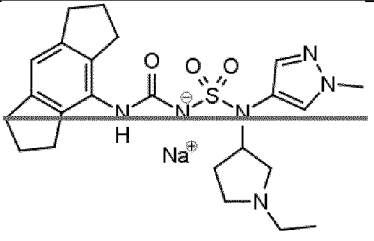
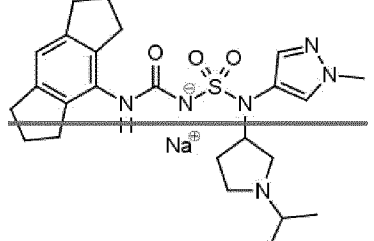
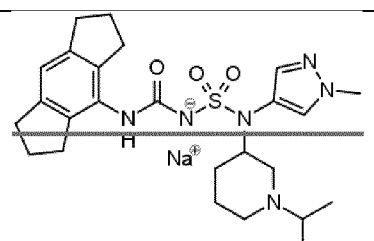
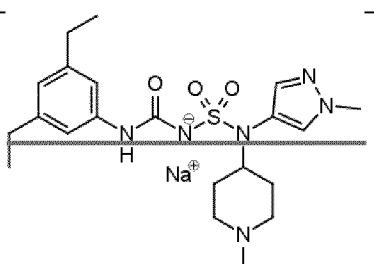
Compound No.	Structure
20	 <chem>CN1C=NC=C1N(C2CCN(C2)CC)S(=O)(=O)NC(=O)N3C4=CC=CC5C4(CCC5)C3.[Na+]</chem>
21	 <chem>CN1C=NC=C1N(C2CCOCC2)S(=O)(=O)NC(=O)N3C4=CC=CC5C4(CCC5)C3.[Na+]</chem>
22	 <chem>CN1C=NC=C1N(C2CCOCC2)S(=O)(=O)NC(=O)N3C4=CC=CC5C4(CCC5)C3.[Na+]</chem>
23	 <chem>CN1C=NC=C1N(C2CCN(C2)CC)S(=O)(=O)NC(=O)N3C4=CC=CC5C4(CCC5)C3.[Na+]</chem>

Compound No.	Structure
24	 <chem>CN1CCN(C1)C2=CN=C(C=C2)N(S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C4C3)C6=CC=CC=C6</chem>
25	 <chem>Fc1cc(F)n(c1)N(S(=O)(=O)N2C(=O)N3C4=CC=CC=C4C3C2)C5=CC=CC=C5</chem>
26	 <chem>FC(F)(F)CN1=CN=C(C=C1)N(S(=O)(=O)N2C(=O)N3C4=CC=CC=C4C3C2)C5=CC=CC=C5</chem>
27	 <chem>COC1=CN=C(C=C1)N(S(=O)(=O)N2C(=O)N3C4=CC=CC=C4C3C2)C5=CC=CC=C5</chem>
28	 <chem>CN1CCN(C1)C2=CN=C(C=C2)N(S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C4C3)C6=CC=CC=C6</chem>
29	 <chem>COC1=CN=C(C=C1)N(S(=O)(=O)N2C(=O)N3C4=CC=CC=C4C3C2)C5=CC=CC=C5</chem>

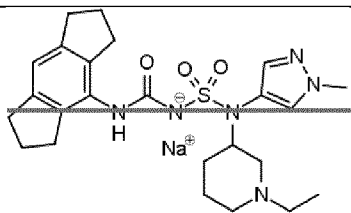
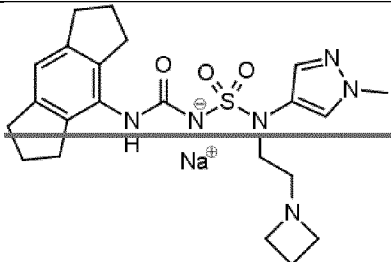
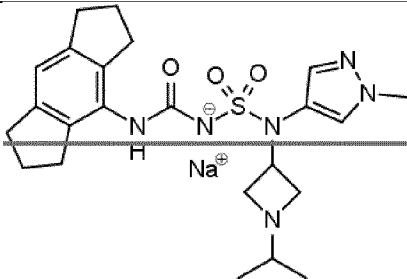
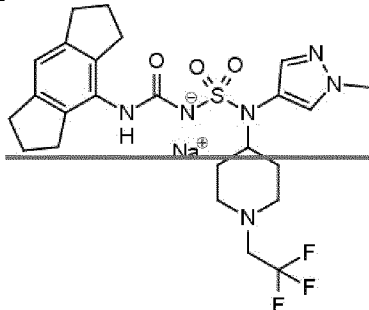
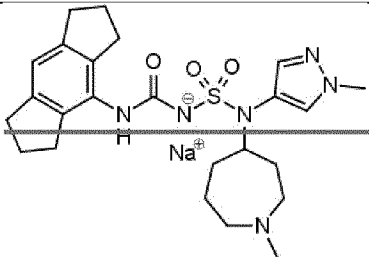
Compound No.	Structure
30	
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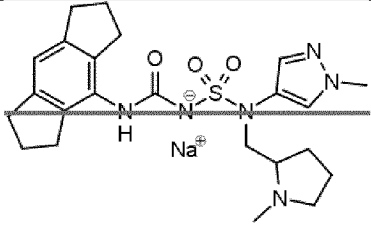
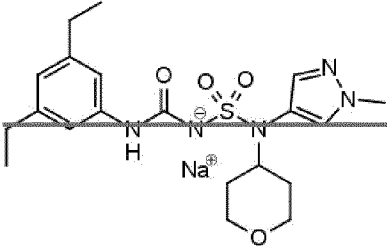
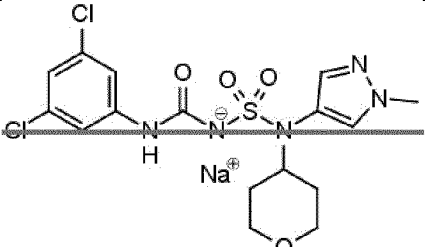
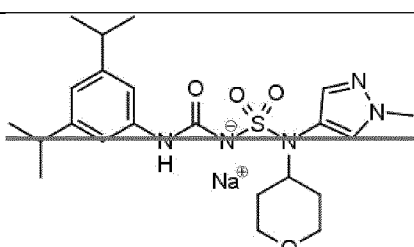
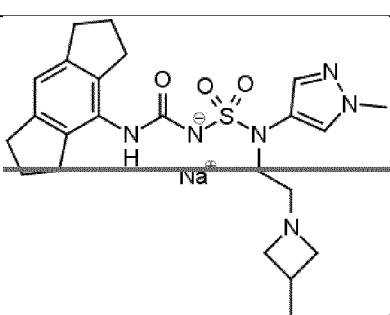
Compound No.	Structure
36	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>
37	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>
38	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>
39	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>
40	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>
41	 <chem>CC1CCN(CC1)C2=CC=C(C=C2)S(=O)(=O)N3C(=O)N4C5=CC=CC=C5C6C7C=CC(=C6)C8C9C=CC(=C8)C9C(=O)N4C3.[Na+]</chem>

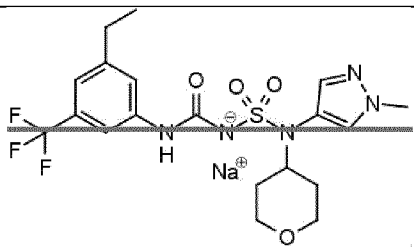
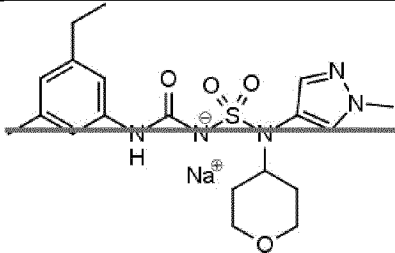
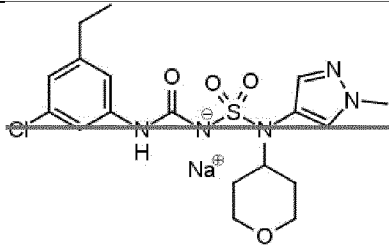
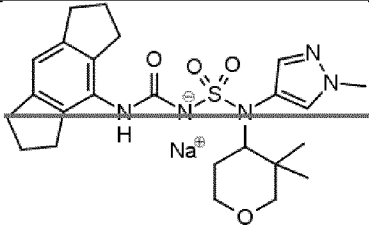
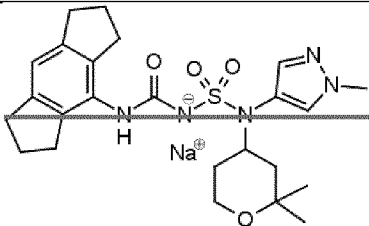
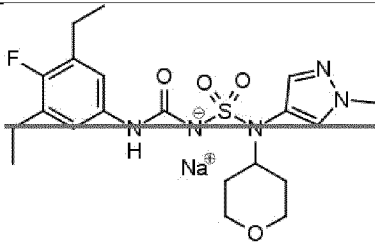
Compound No.	Structure
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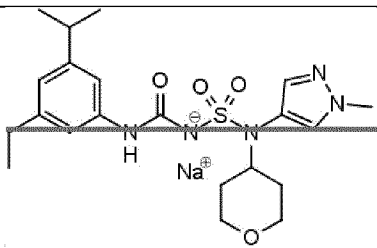
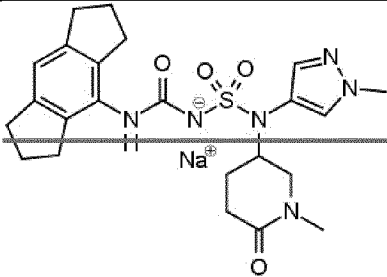
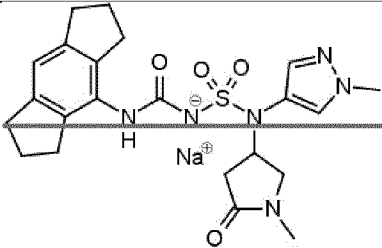
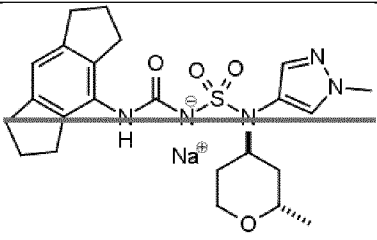
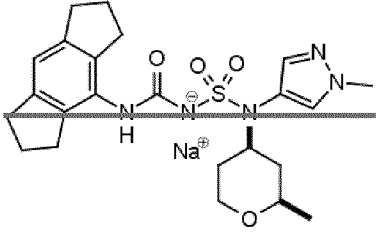
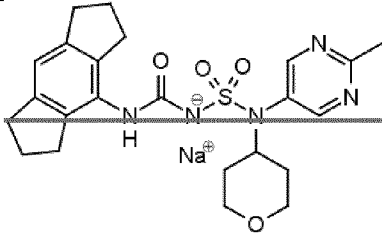
Compound No.	Structure
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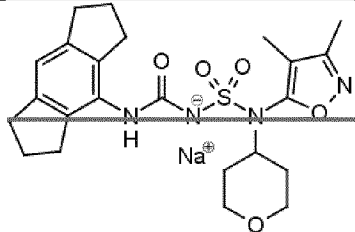
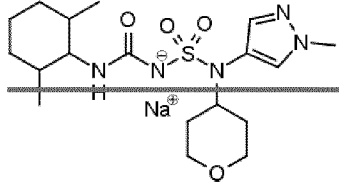
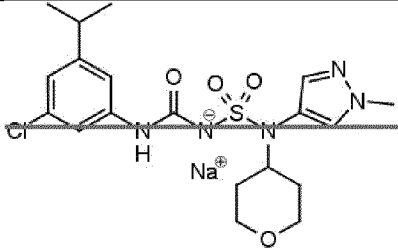
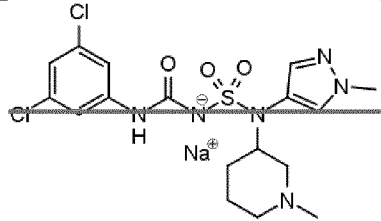
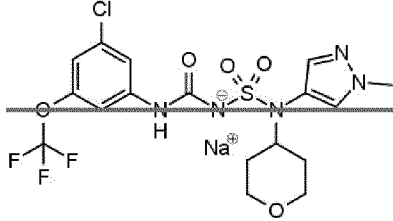
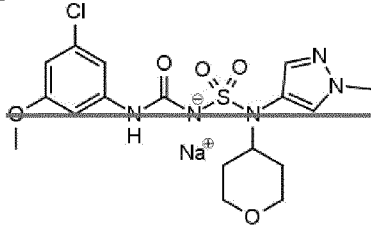
Compound No.	Structure
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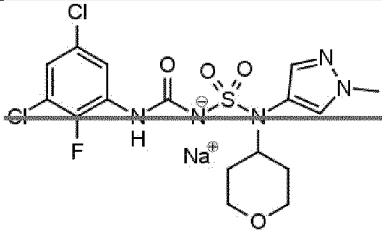
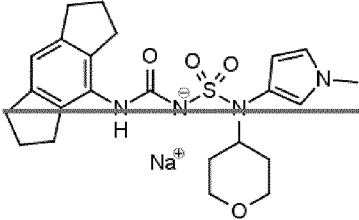
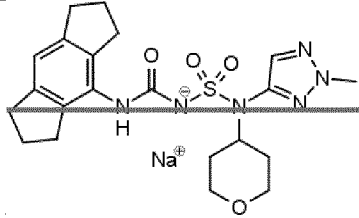
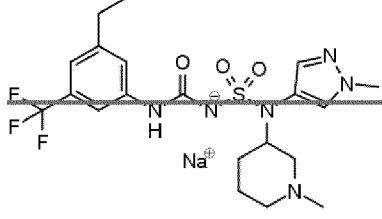
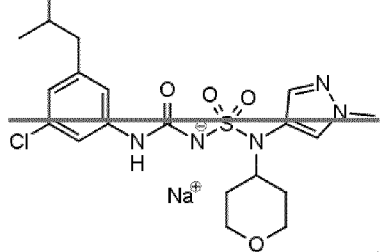
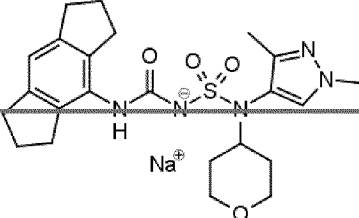
Compound No.	Structure
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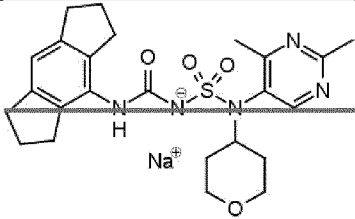
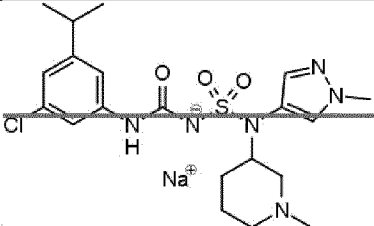
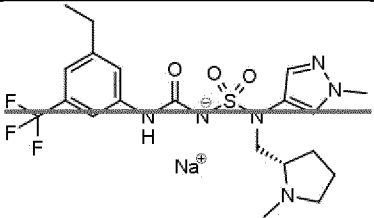
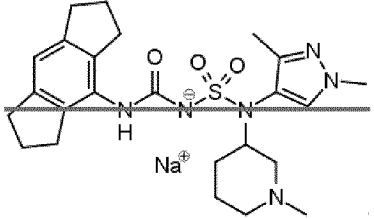
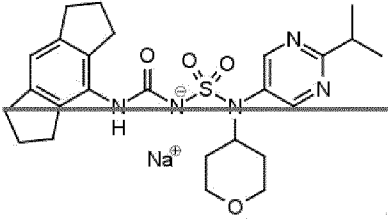
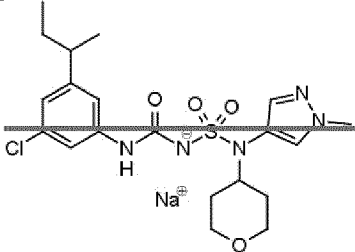
Compound No.	Structure
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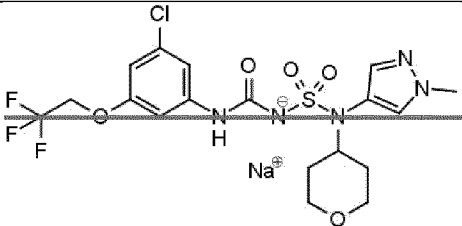
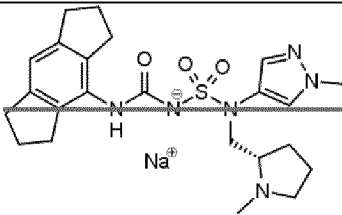
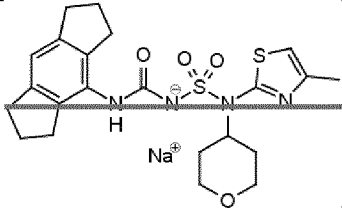
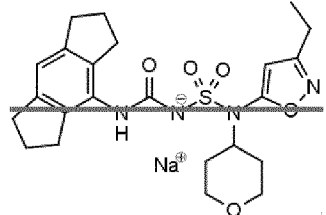
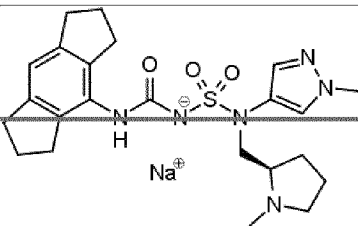
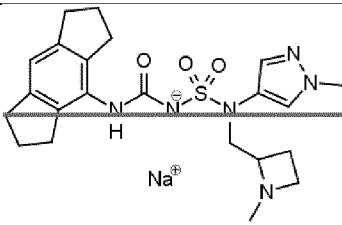
Compound No.	Structure
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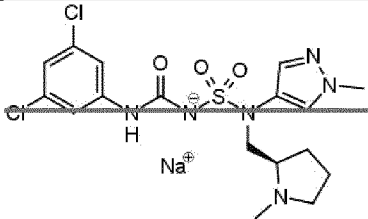
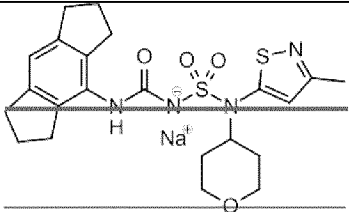
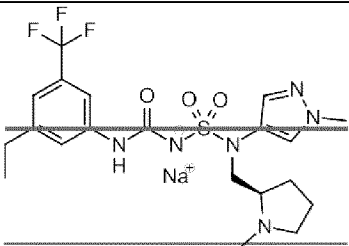
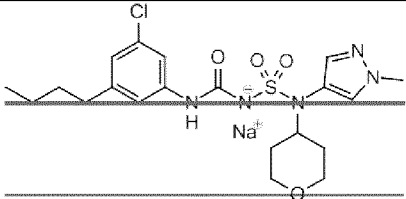
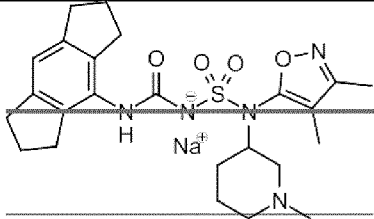
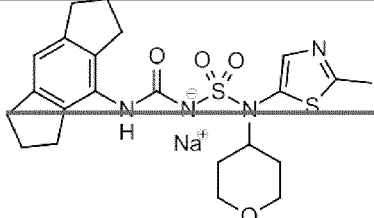
Compound No.	Structure
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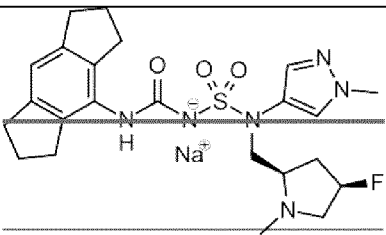
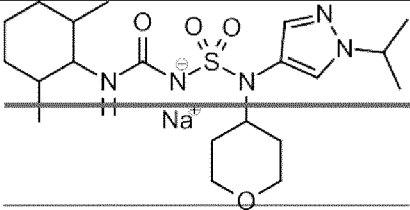
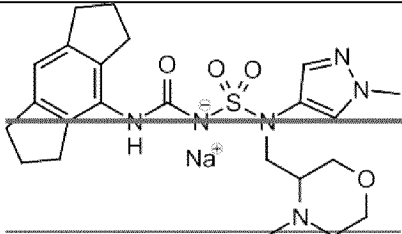
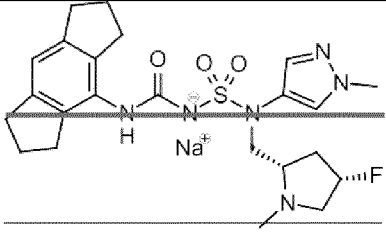
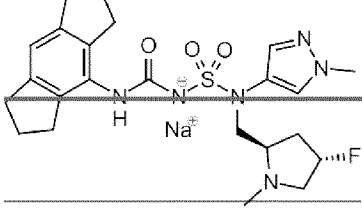
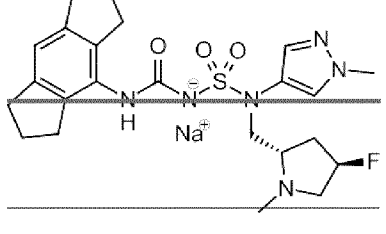
Compound No.	Structure
87	
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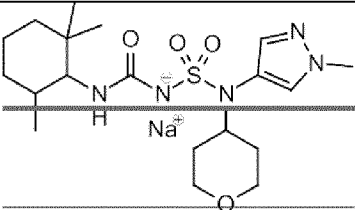
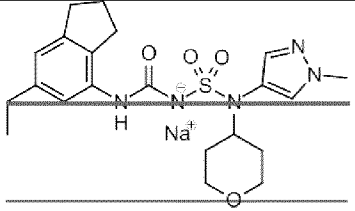
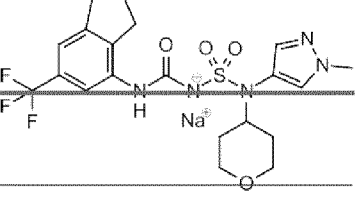
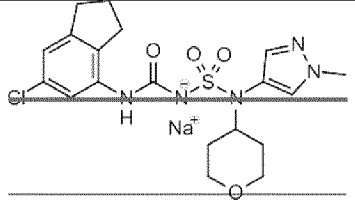
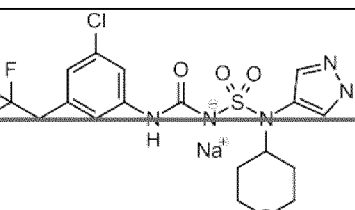
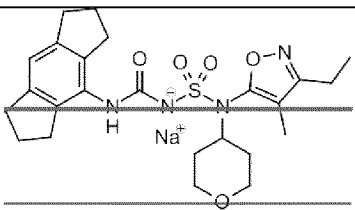
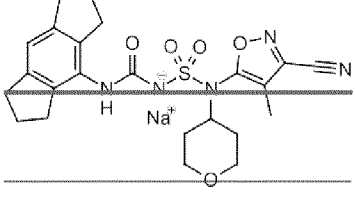
Compound No.	Structure
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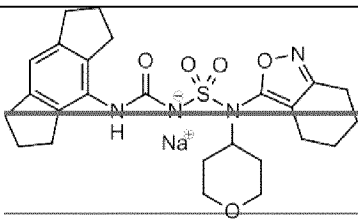
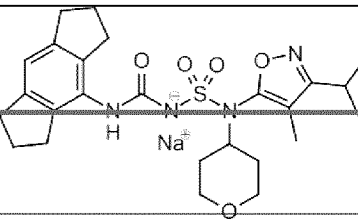
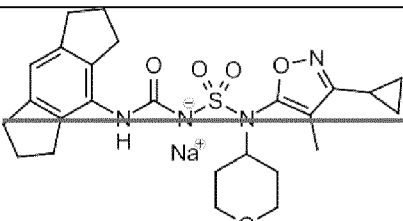
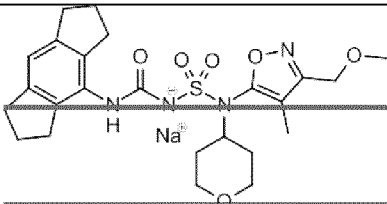
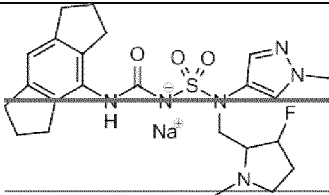
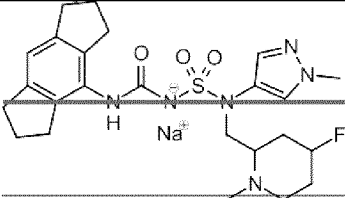
Compound No.	Structure
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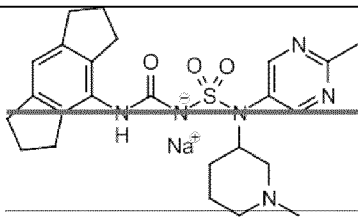
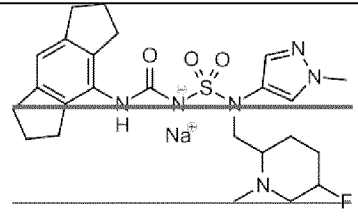
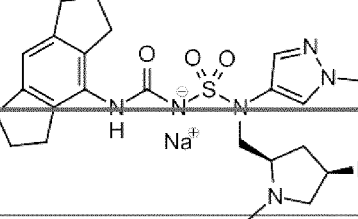
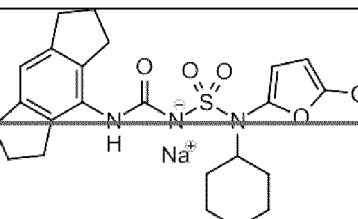
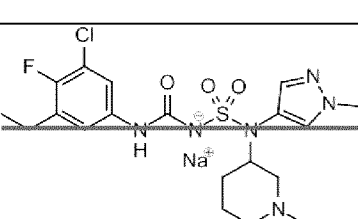
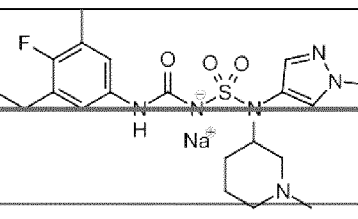
Compound No.	Structure
106	
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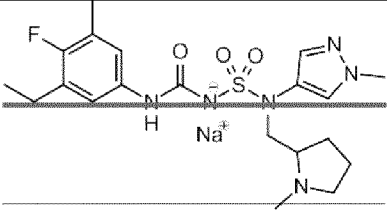
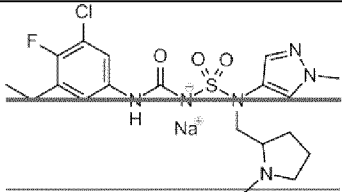
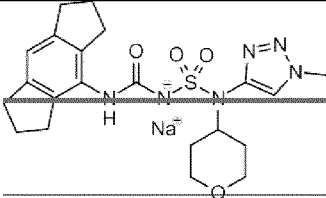
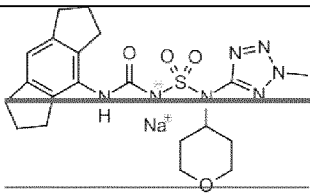
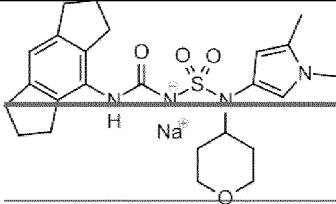
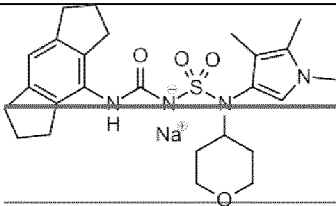
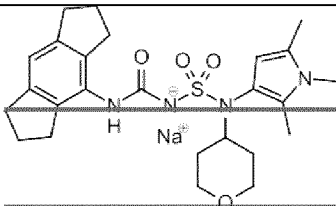
Compound No.	Structure
112	
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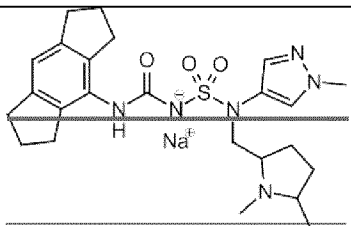
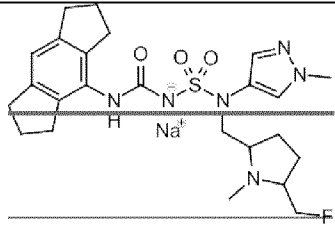
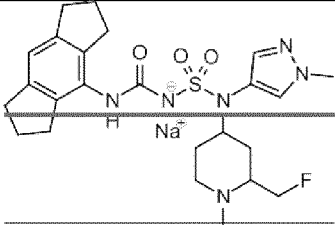
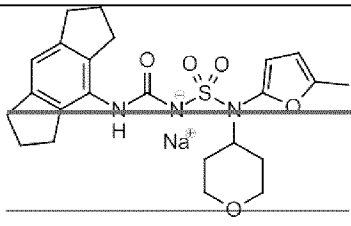
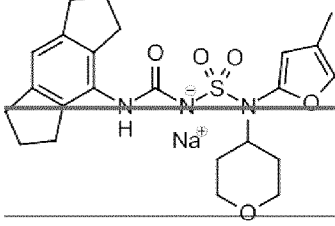
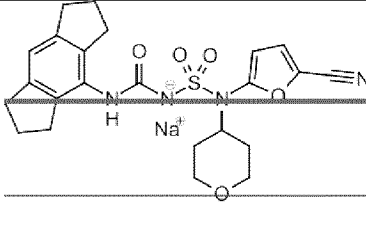
Compound No.	Structure
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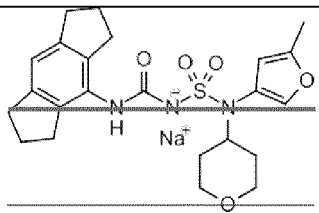
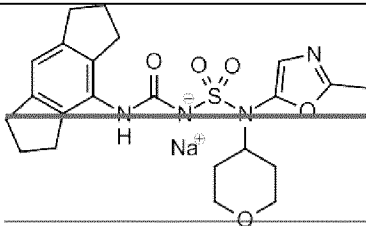
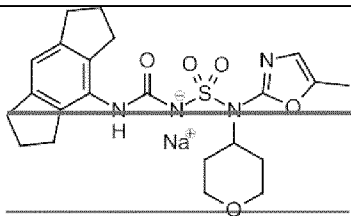
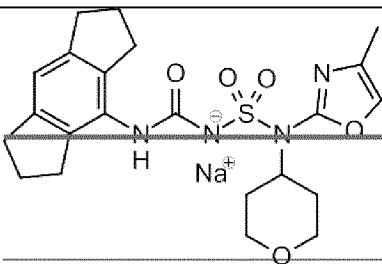
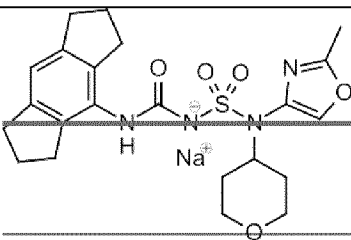
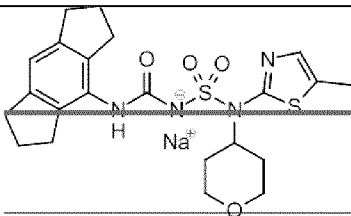
Compound No.	Structure
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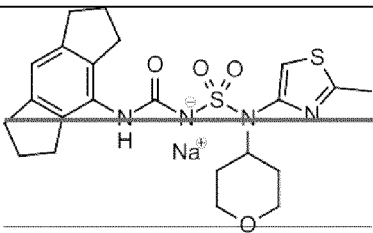
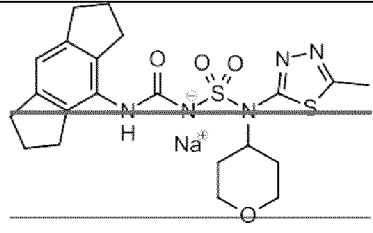
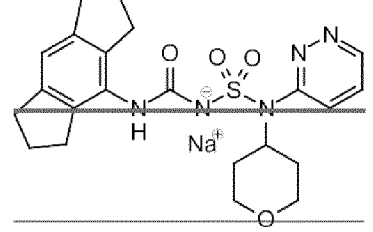
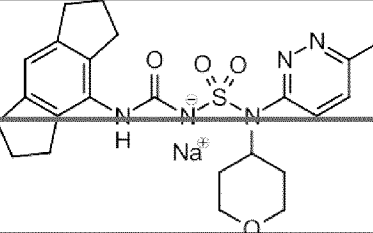
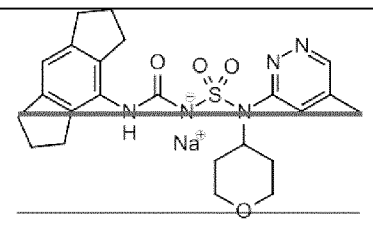
Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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and pharmaceutically acceptable salts thereof.

106. A pharmaceutical composition comprising the compound of any one of claims 1 to 95 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

117. The compound of any one of claims 1 to 95, or the pharmaceutical composition of claim 106,

for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

~~128~~. The compound of any one of claims 1 to ~~95~~, or the pharmaceutical composition of claim ~~106~~, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

~~139~~. The compound or pharmaceutical composition for use of claim ~~117~~ or claim 128, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

~~1410~~. The compound or pharmaceutical composition for use of claims ~~117~~ or claim 128, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

~~1511~~. The compound or pharmaceutical composition for use of claims ~~117~~ or claim 128, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.